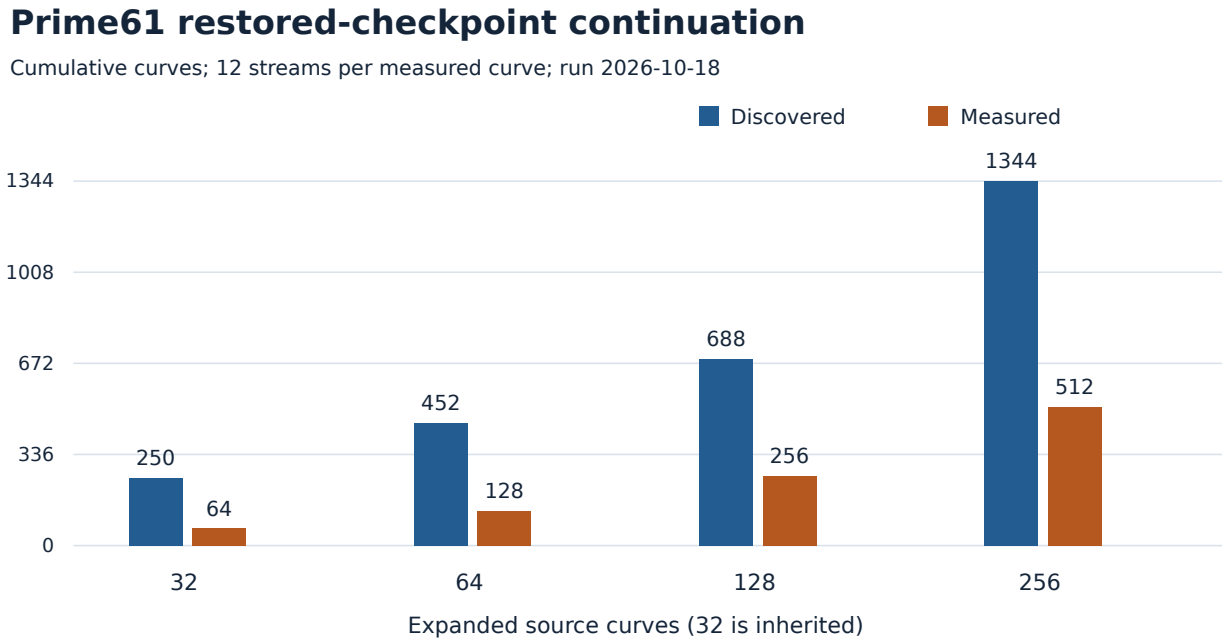
Prime61: continuation from a Drive checkpoint

2026-10-18 · capacity and relation-stage observations

The public synthetic experiment over $p = 2^{61} - 1$ continues from the verified 2026-10-17 checkpoint. The table contains cumulative populations; the first row was inherited. These are finite prefixes of the selected-prime component. The entire horizontal class has not been enumerated or certified.

Expanded	Discovered	Maps	Measured	Streams
32	250	480	64	768
64	452	960	128	1536
128	688	1920	256	3072
256	1344	3840	512	6144



Source: results/runs/{restore-verify,verify-64,verify-128,verify-256}.stdout

Exact population counts; no statistical interval. Selected-prime frontier remains open.

New since the restored checkpoint: 1094 discovered curves, 3360 map records, 5376 committed collection streams and 537600000 target trials. The final population has 0 relations and 0 verified solves; 1088 discovered sources remain pending. Every measured curve has twelve capped streams. A capped stream is retained as a negative result.

The last live-database status records 3387392 bytes. The maximum reported native-worker peak RSS across the retained phase receipts is 18396 KiB. Compiler, test, audit and renderer processes are excluded from this worker statistic. These are capacity observations on this prefix, with no full-class memory extrapolation or wall-speed claim. Full IC cost and speedup remain unknown; no matched rho or external-tool speed comparison was performed.

Inputs, controls and remaining obligations

The source is the exact previously validated revision `b68341cd72b51acf9c5ce6c4f755351cc83053db`, built with Rust 1.99.1. The initial current-main build failed in unrelated ecbench code; `AMENDMENT.json` froze the fallback before new collection. Both source revisions' build/test outcomes are retained and distinguished in the README.

The unchanged spec fixes $a = p - 3$, $b = 111$, prime order `2305843010818082053`, `G=(1,1796364306280342617)` and `Q=(992726311114892094,558260110083040833)`. The public checking scalar is 539365399; the collector receives G and Q. Prime degrees are 3, 7, 11, 13, 19, 23, 53 and 61. Each curve uses a fresh low-x base with three inverse pairs and requests twenty relations. The twelve seeds are 2026101600 through 2026101611, each with 100000 target trials. One first-parent transported workload is measured per curve.

The downloaded archive's SHA-256 and all internal manifest entries passed. Before continuation, the native verifier checked the restored maps and rosters, replayed the first sixteen curves' 192 collector streams, and reproduced the archived logical dump exactly. The final immutable-prefix audit checks all inherited payloads. Every batch receives complete native map, source-roster, point and record verification; the optional collector replay is the separate restored-control population, not extra headline runs.

Raw phase commands, stdout, stderr, exits and snapshots are preserved in `results/runs.tar.xz`; unpack from the study directory with `tar -xJf results/runs.tar.xz -C results`. Compact verification, prefix and storage receipts sit under `results/`. The immutable final Drive archive is separately downloaded and checked before its receipt is called passing. Private Drive object identifiers are omitted from this public report.

The earlier fundamental-discriminant result excludes vertical separable base-field edges for this fixed ordinary trace class. No new vertical-map or algebraic-identity theorem is claimed. Exhaustive class structure, generation by the chosen degree set, every class workload and full-run storage/compute provisioning remain open. A merge or passing prefix does not discharge those obligations. Candidate registry promotion is pending; full EC1/UIDs and map witnesses stay in the raw logical records.

Evidence and reproduction: `PROTOCOL.json`, `AMENDMENT.json`, `spec.json`, `continue.sh`, `audit_prefix.rs`, `visual.rs`, `reports/summary.json`, and the study README. The SVG is generated only from native phase status records. The historical pilot and performance dashboards are unchanged: this run adds coverage observations without an admitted performance result.

Source-linked study and raw-evidence manifest